

AMBROSE LIGHT, NY, USA

WMO: 994100

Lat: 40.460N

Lon: 73.830W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	16.9	N/A	N/A	N/A	N/A	N/A	N/A	47.6	28.4	43.8	32.9	24.5	320	0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	84.1	N/A	81.0	N/A	78.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	15.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	43.9	38.7	34.3	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	33.4	33.9	40.0	49.9	57.6	67.8	73.1	72.7	67.6	58.3	48.4	39.7
	DBStd	15.54	10.08	8.20	6.92	6.73	5.79	4.97	4.00	4.10	4.05	5.96	7.17	7.20
	HDD50	1814	519	453	318	80	3	0	0	0	0	6	112	323
	HDD65	4878	981	871	776	457	243	27	0	1	21	221	497	783
	CDD50	3140	3	2	6	76	237	534	716	703	529	264	65	5
	CDD65	729	0	0	0	3	12	110	251	239	100	14	0	0
	CDH74	2211	0	0	0	15	52	315	896	809	99	25	0	0
	CDH80	359	0	0	0	2	12	56	159	122	6	2	0	0
Wind	WSAvg	17.7	20.6	19.7	18.7	19.8	17.6	15.3	14.0	13.8	15.6	17.3	19.2	20.3
Precipitation	PrecAvg	45.8	3.5	3.0	4.2	3.9	3.8	3.9	3.7	4.5	3.8	3.9	3.4	4.3
	PrecMax	57.4	8.1	5.8	8.8	8.7	8.0	9.0	6.5	13.4	9.0	15.2	5.8	7.4
	PrecMin	36.5	1.6	1.0	1.1	1.6	1.3	0.9	1.0	0.7	1.4	0.3	1.3	1.3
	PrecStd	5.7	1.3	1.3	1.8	1.7	1.4	2.0	1.5	2.5	1.9	2.9	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	59.2	63.6	76.1	79.4	85.6	88.2	88.8	80.4	77.2	66.3	57.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	53.0	50.8	55.3	68.6	73.0	80.8	84.3	83.2	76.8	71.7	62.9	54.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	50.0	47.1	51.0	63.5	68.5	77.6	81.1	80.1	74.7	68.7	60.4	52.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	46.8	43.9	48.0	58.7	65.5	74.4	78.3	78.1	73.2	66.8	58.8	49.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	9.9	9.7	9.3	10.1	9.1	8.4	7.0	6.4	6.6	8.0	8.5	9.0
		MCDBR	12.4	13.0	14.3	17.8	14.9	13.9	11.5	9.7	8.2	9.8	10.6	11.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.321	0.331	0.361	0.385	0.431	0.485	0.490	0.476	0.402	0.365	0.335	0.320	
	taud	2.479	2.456	2.413	2.378	2.281	2.170	2.168	2.199	2.379	2.485	2.524	2.529	
	Ebn at Noon	266	281	282	282	270	254	252	252	265	265	259	258	
	Edh at Noon	24	29	34	37	42	47	47	44	35	28	23	22	
All-Sky Solar Radiation	RadAvg	597	886	1204	1548	1719	1892	1914	1687	1390	975	682	517	
	RadStd	51	91	103	148	178	175	92	109	124	94	59	47	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (62 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+122

Nomenclature: See separate page